

数学作业纸

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$$2.3.7 \quad V_{01} = V_i \left(1 + \frac{40k\Omega}{12k\Omega} \right) = 2.167 \text{ mV}$$

$$V_{02} = -\frac{30}{12} V_i = -1.25 \text{ mV}$$

$$V_0 = V_{01} - V_{02} = 3.417 \text{ mV}$$

$$A_v = 6.834$$

$$R_1 = R_{11} // R_{12} = 10k\Omega // 12k\Omega = 6.834k\Omega$$

$$2.3.8 \quad V_C = V_{B1} = V_{B2} = 6V$$

$$V_{B3} = V_{B2} = 0V$$

$$V_E = V_{B3} - V_{B2} = 0.17V$$

$$(2) \quad \bar{I}_B = \frac{V_{01}}{R_2} = \frac{200mV}{10k\Omega} = 20\mu A$$

$$\beta = \frac{I_C}{I_B} = 50$$

$$I_C = \frac{V_1 - V_{B1}}{R_1} = \frac{12V - 6V}{6k\Omega} = 1mA$$

$$2.3.11 \quad \frac{V_{REF} - V_A}{R_1} = \frac{V_A - V_{01}}{R(1+\beta)}$$

$$R(1+\beta)V_{01} = -\frac{V_{REF}}{R_1} R(1+\beta)$$

$$\frac{V_0}{R_2} = \frac{V_{REF}}{R_1} + \frac{V_{01}}{R}$$

$$V_0 = -\frac{V_{REF} R_2}{R_1} - \frac{V_{01} R_2}{R} = -\frac{V_{REF}}{R_1} R_2 + \frac{V_{REF}}{R_1} (1+\beta) R_2$$

$$= \frac{R_2 V_{REF}}{R_1} \beta$$

V_0 与 β 呈线性关系

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$$2.4.5 \cdot \frac{-V_1}{R_2} = \frac{V_1 - V_0}{R_1} + \frac{V_1 - V_2}{R_G}$$

$$\frac{V_2 - V_0}{R_2} = \frac{V_1 - V_2}{R_G} + \frac{V_0 - V_2}{R_1} \Rightarrow V_0 = - \left(1 + \frac{R_2}{R_1} + \frac{2R_2}{R_G} \right) (V_1 - V_2)$$

$$2.4.7 \cdot \frac{V_{i1} - V_n}{R_1} + \frac{V_{i2} - V_n}{R_2} = \frac{V_n - V_0}{R_6} \quad V_n = V_p$$

$$\frac{V_{i3} - V_p}{R_3} + \frac{V_{i4} - V_p}{R_4} = \frac{V_p}{R_5}$$

$$V_0 = -\frac{5}{4} V_{i1} - 2V_{i2} + \frac{51}{22} V_{i3} + \frac{51}{44} V_{i4}$$

$$2.4.12 \cdot V_{o1} = V_{i1} \quad V_{o2} = V_{i2}$$

$$\frac{V_{o1} - V_{i3}}{R_1} = \frac{V_{i3}}{R_2} \quad V_{i3} = V_{p1}$$

$$\frac{V_{o2} - V_{p3}}{R_1} = \frac{V_{p3} - V_{o4}}{R_2}$$

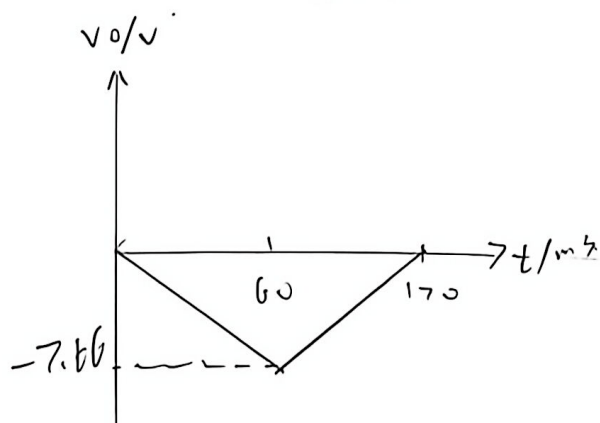
$$V_0 = -\frac{R_4}{R_3} V_{o4}$$

$$\Rightarrow \frac{V_0}{V_{i1} - V_{i2}} = -\frac{R_4 R_2}{R_3 R_1}$$

$$2.4.13. (1) V_0 = -\frac{1}{RC} \int_0^1 V_1 dt = -5V$$

$$(2) V_0(t_1) = -\frac{1}{RC} \int_0^{60ms} V_1 dt = -7.66V$$

$$V_0(t_2) = -\frac{1}{RC} \int_{60ms}^{170ms} V_2 dt + V_0(t_1) = 0V$$



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$$2.4.14. (1) \quad V_{01} = \left(\frac{R_2}{R_2 + R_1} \right) \left(1 + \frac{R_4}{R_1} \right) V_{I2} - \frac{R_4}{R_1} V_{I1}$$

$$\textcircled{2} \quad V_{02} = -\frac{1}{R_5} \int_0^t \left(\frac{V_{I1}}{R_5} + \frac{V_{I2}}{R_6} \right) dt$$

$$= -\frac{1}{C} \int_0^t \left[\frac{1}{R_5} \left(\frac{R_2}{R_2 + R_1} \right) \left(1 + \frac{R_4}{R_1} \right) V_{I2} - \frac{R_4}{R_5 R_1} V_{I1} + \frac{V_{I2}}{R_6} \right] dt$$

$$\textcircled{3} \quad V_{02} = -\frac{1}{R_5 C} \int_0^t \left[(V_{I2} - V_{I1}) + V_{I2} \right] dt$$

$$2.4.15 (1) \quad V_P(s) = \frac{\frac{1}{sC}}{R_2 + \frac{1}{sC}} V_{I2}(s)$$

$$V_{01}(s) = \left[1 + \frac{sC}{R_2} \right] V_P(s) = \frac{1}{sR_2C} V_{I2}(s)$$

$$V_{01} = \frac{1}{R_2 C} \int_0^t V_{I2} dt$$

$$(2) \quad V_{01} = -\frac{1}{sR_2C} V_{I2}(s)$$

$$V_{02} = -\frac{1}{R_2 C} \int_0^t V_{I2} dt$$

$$(3) \quad V_{02} = V_{01} + V_{02} = \frac{1}{R_2 C} \int_0^t (V_{I2} - V_{I1}) dt$$